Galaxy ProTo 2.0 Redux

~ Dynamic Tool Prototyping using Interactive Tools

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As one of many possible web servers implementing the proposed " Interactive client tool " idea, we present:

Galaxy ProTo 2.0 Redux

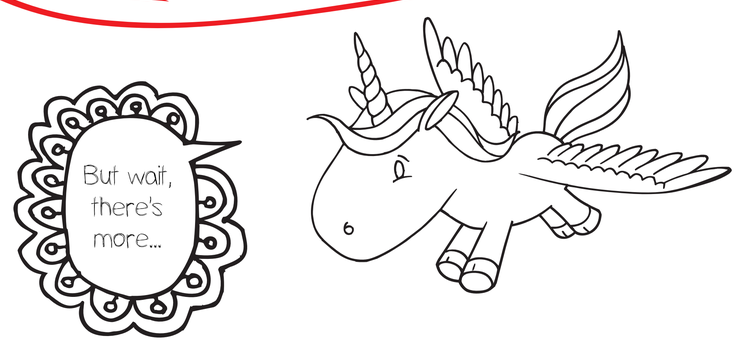
Short for " Galaxy Prototyping Tool API "

Tool form defined in pure python™

Fully dynamic generation of tool GUI from the return value of python methods. Some examples:

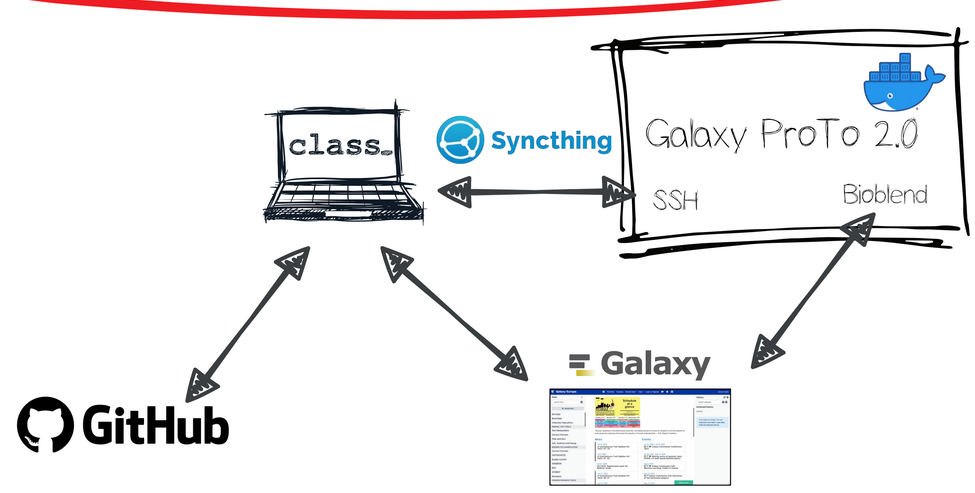
`return true`	→	Check box
`return ['Selection A', 'Selection B']`	→	Selection box: Selection A
`return 'Default text'`	→	Text area: Default text
`return { 'Something': True, 'Something else': False }`	→	Check box list: Check all / Uncheck all; Something; Something else
`return '__history__'`	→	History selection box: 1: Pasted Entry [hg38]

All GUI elements can be dynamically generated based on previous user selections, databases, dataset contents, computations -- really anything!



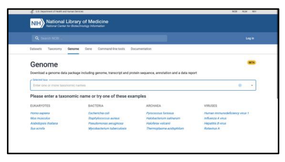
Using Galaxy as an Integrated Development Environment (IDE)!

A developer can code the tool GUI and job execution functionality " on the fly " on any Galaxy server just as a regular user!



Three types of tools in Galaxy
(well actually 33, but who's counting?)

Data source tools



Opens new window with external page

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Returns search results through a dataset import job. No data flow in the other direction

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Any GUI possible, but requires deployment of Galaxy-specific functionality at remotely hosted site

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Not rerunnable / workflowable

Interactive tools

Spins up a Docker container and connects to it in a new window

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Uses API calls to get/put datasets

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Anything goes in terms of GUI and functionality

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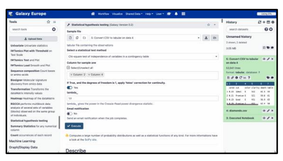
Rerunnable / workflowable (sort of), but all processing happens inside the container

Regular tools



GUI opens up in the middle panel

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Fully integrated in Galaxy with all " bells and whistles " related to dataset input / output

Based on Galaxy tool form XML, which is extensive, but still limited

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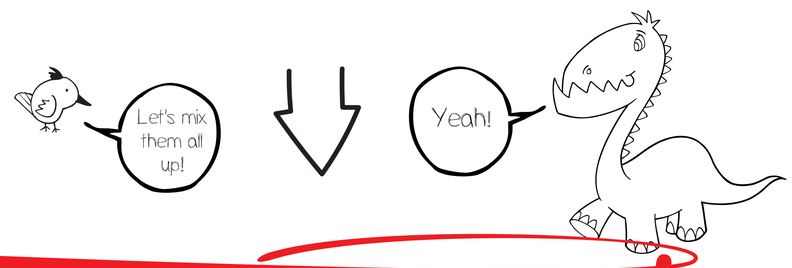
Fully rerunnable / workflowable, jobs are sent to the job runner



One new tool type with two modes
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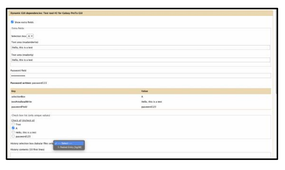
Refurbished interactive tool

" Data source tool " mode



Operates as a data source tool showing content in the middle panel

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Connects to an internal web server running as a separate interactive tool job

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No external web server needed

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No waiting time needed to get to the GUI if the interactive (server) tool is already running (it is just a browser redirect)

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Anything goes in terms of GUI and functionality

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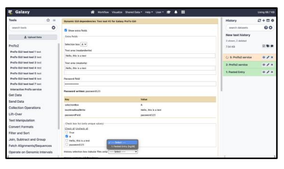
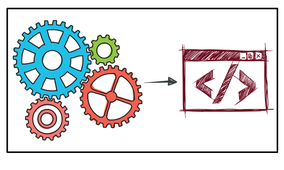
Interactive client tool
(or a better name, if you have one?)



Interactive tool



Spins up a Docker container only for the first access

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Web server that only serves the GUI

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Uses API calls to get/put datasets

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" Regular tool " mode

Jobs are executed through the Galaxy job runner

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Jobs are containerized using the same or a different image as the interactive tool

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Fully rerunnable / workflowable

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Clicking rerun opens up the tool in " data source " mode, sending the job parameters to the interactive tool web server which regenerates the GUI

SSH access directly to the interactive tool container!

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Use Syncthing through an SSH tunnel to automatically sync the code from your laptop to the container

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Galaxy ProTo 2.0 allows realtime GUI changes

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Execute long-running jobs directly using the Galaxy backend

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Perfect for letting other researcher try out the functionality on real data with extremely fast turnaround for bugfixes etc.

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Galaxy rerun functionality is optimal for bugfixing

Run container in user space (e.g. Podman)

Root -> galaxy user in container
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Limit mounted volumes